Electric Circuits

ELEC 221 – Electrical Circuits

Alireza Bakhshai, PhD

Learning Outcomes

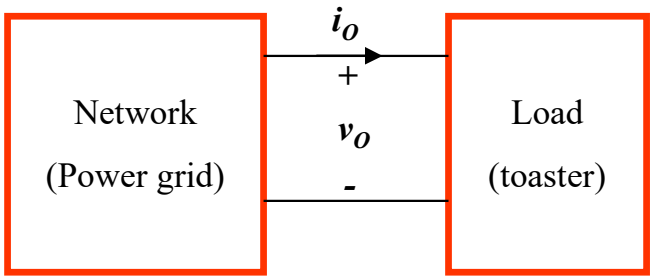
By the end of lecture, you will be able to learn and use:

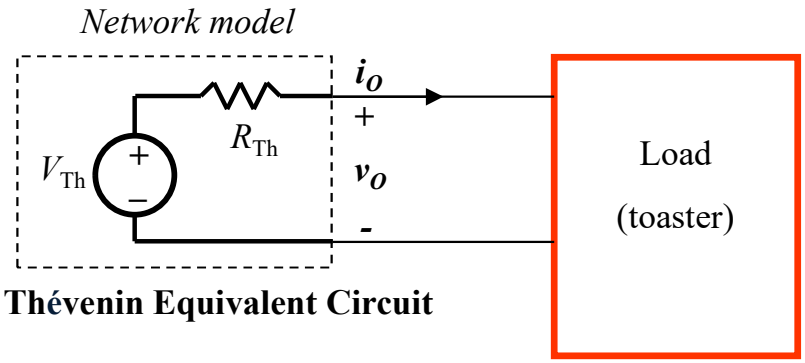
- Thévenin and Norton Equivalent circuits and transformations.
- Maximum Power Transfer
- Principle of superposition

Thévenin and Norton Equivalent Circuits

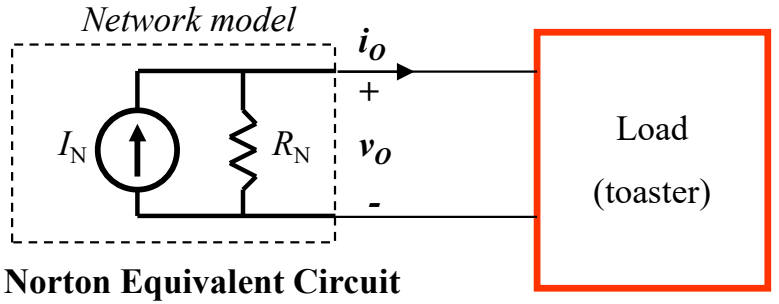
In many practical situations we are only interested in the voltage and current of the two terminals of a network as the load changes:

Example: A toaster connected to the power outlet.





Thévenin Equivalent Circuit

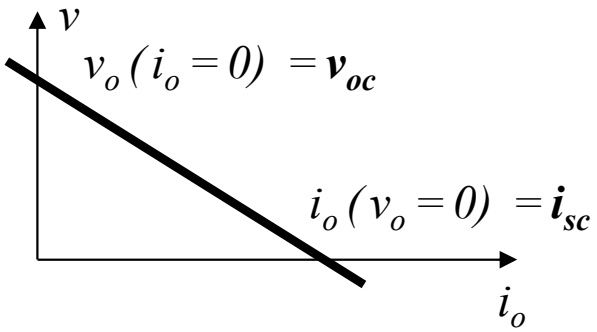


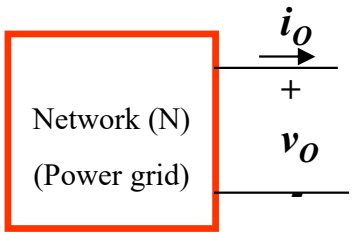
Norton Equivalent Circuit

The goal is to model and replace the network "N" with a simple equivalent (**Thévenin** or **Norton**) circuit and then proceed with the analysis.

Thévenin Theorem

Purely resistive circuit: Network "N" is composed of independent and/or dependent sources and resistors only . No assumption is made about the load.





Thévenin Theorm: The network "N" can be replaced by an equivalent circuit that contains only an independent voltage source in series with a resistor :

$$\begin{cases} V_{Th} = v_{oc} \\ R_{Th} = \dfrac{V_{Th}}{i_{sc}} = \dfrac{v_{oc}}{i_{sc}} \end{cases}$$

$v_0 = v_{Th} - R_{Th}\, i_0$

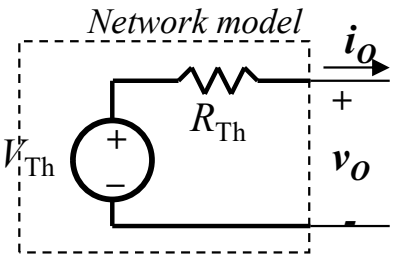
$i_0 = 0$, open circuit

$\Rightarrow v_0 = v_{oc} = V_{Th}$

$v_o = 0$, short circuit

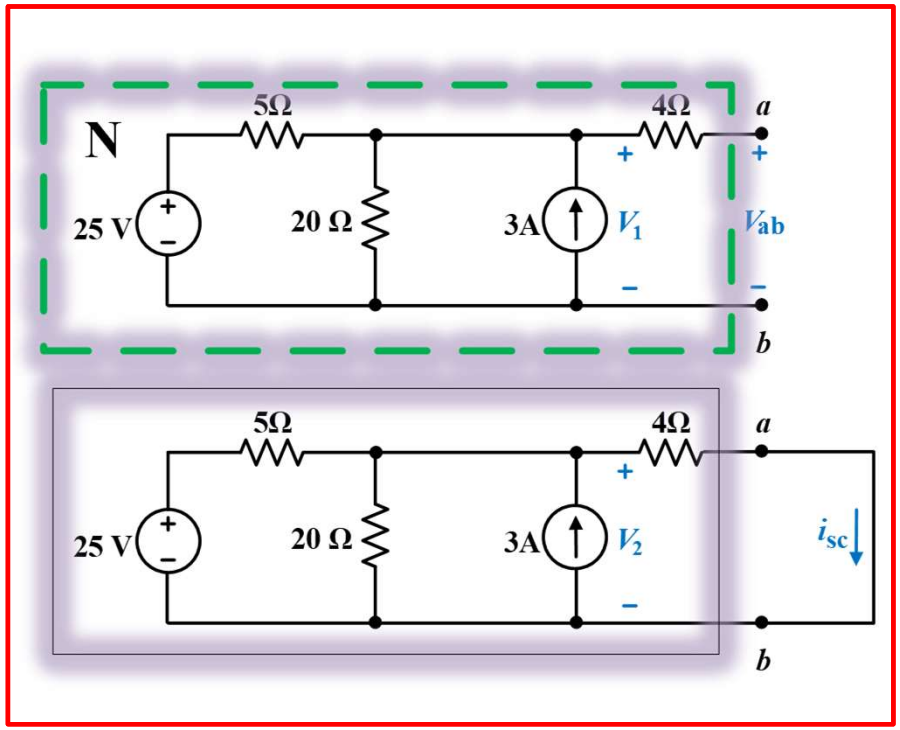
$\Rightarrow V_{Th} - R_{Th}\, i_{sc} = 0$

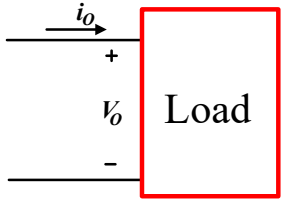
$\Rightarrow i_0 = i_{sc} = V_{Th}/R_{Th}$



Example: Thévenin Equivalent Circuits

- Find the Thévenin equivalent of circuit "N".





Open circuit: $v_{oc} = ?$

Use node voltage method

$$\frac{v_1 - 25}{5} + \frac{v_1}{20} - 3 = 0$$

$$\Rightarrow v_1 = 32 \text{ V}$$

$$\Rightarrow v_{Th} = v_{oc} = v_1 = 32 \text{ V}$$

Short circuit: $i_{sc} = ?$

Again, use node voltage method:

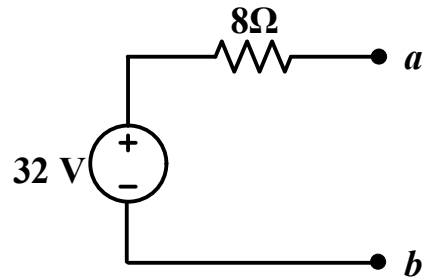
$$\frac{v_2 - 25}{5} + \frac{v_2}{20} - 3 + \frac{v_2}{4} = 0$$

$$\Rightarrow v_2 = 16 \text{ V} \rightarrow i_{sc} = v_2/4 = 4 \text{ A}$$

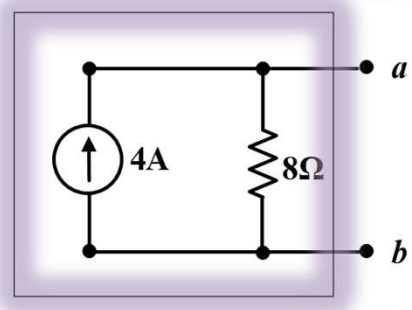
Example: Thévenin Equivalent Circuits

- Find the Thévenin equivalent of circuit "N".

$R_{Th} = V_{Th} / i_{sc}$
$= 32/4 = 8\ \Omega$



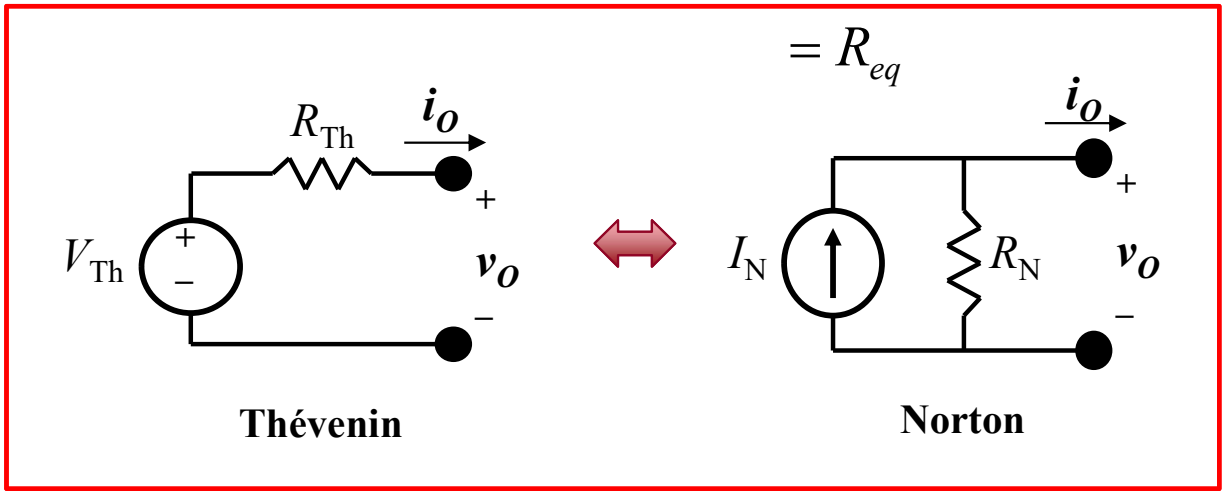
Recall Source transformation:
$I_N = i_{sc} = 4$ A
$R_N = R_{Th} = 8\ \Omega$



Norton Theorem

Norton Theorem: The network "N" can be replaced by an equivalent circuit that contains only an independent current source in parallel with a resistor .

Using source transformation technique and Thévenin equivalent circuit:



$$R_N = R_{Th}$$
$$I_N = V_{Th}/R_{Th} = v_{oc}/R_{Th}$$
$$\Rightarrow I_N = \frac{v_{oc}}{\left(\frac{v_{oc}}{i_{sc}}\right)}$$
$$\boxed{\Rightarrow I_N = i_{sc}}$$

Finding Equivalent Circuits

- To find the Thévenin or Norton Equivalent Circuit for any network we will need **2 out of 3 values**:

$$R_{eq} \qquad i_{sc} \qquad v_{oc}$$

- As:

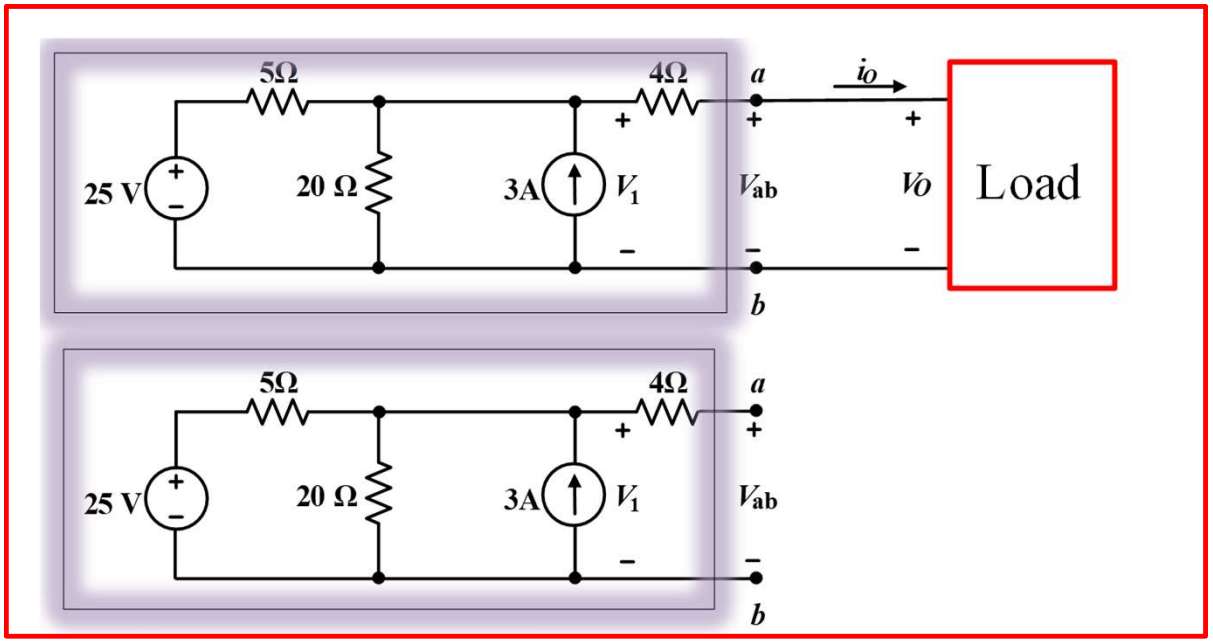
$$v_{oc} = R_{eq} \cdot i_{sc}$$

- There are 4 methods available **in general**:
  - Finding 2 of the above 3 quantities separately: methods 1 to 3, or,
  - Finding 2 of the above 3 quantities simultaneously: method 4!

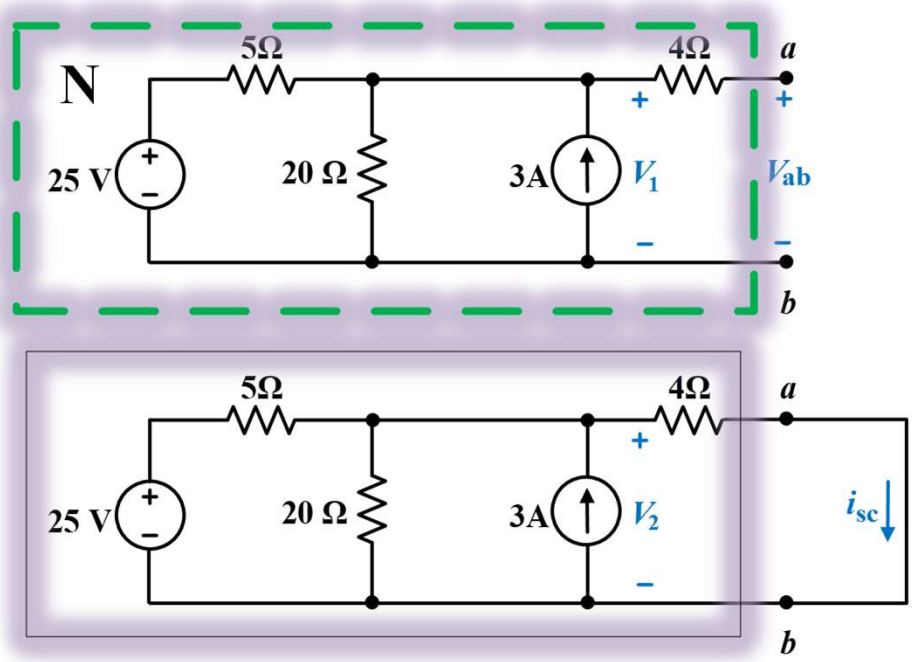
Networks with Only Independent Sources: Method 1→ oc voltage and sc current

Step 1: Detach the network from the load (rest of the circuit).

Step 2: Find the open circuit voltage across the two detached network terminals (v_{oc}): $V_{Th} = v_{oc}$



Networks with Only Independent Sources: Method 1→ oc voltage and sc current



Open circuit: $v_{oc} = ?$

Use node voltage method

$$\frac{v_1 - 25}{5} + \frac{v_1}{20} - 3 = 0$$

$$\Rightarrow v_1 = 32 \text{ V}$$

$$\Rightarrow v_{Th} = v_{oc} = v_1 = 32 \text{ V}$$

Step 3: Short circuit the two detached network terminals and find the current in the short circuit wire (i_{sc}).

$$\frac{v_2 - 25}{5} + \frac{v_2}{20} - 3 + \frac{v_2}{4} = 0$$

$$\Rightarrow v_2 = 16 \text{ V} \rightarrow i_{sc} = v_2/4 = 4 \text{ A}$$

Networks with Only Independent Sources: Method 1→ oc voltage and sc current

Step 4: Calculate the Thévenin resistance:

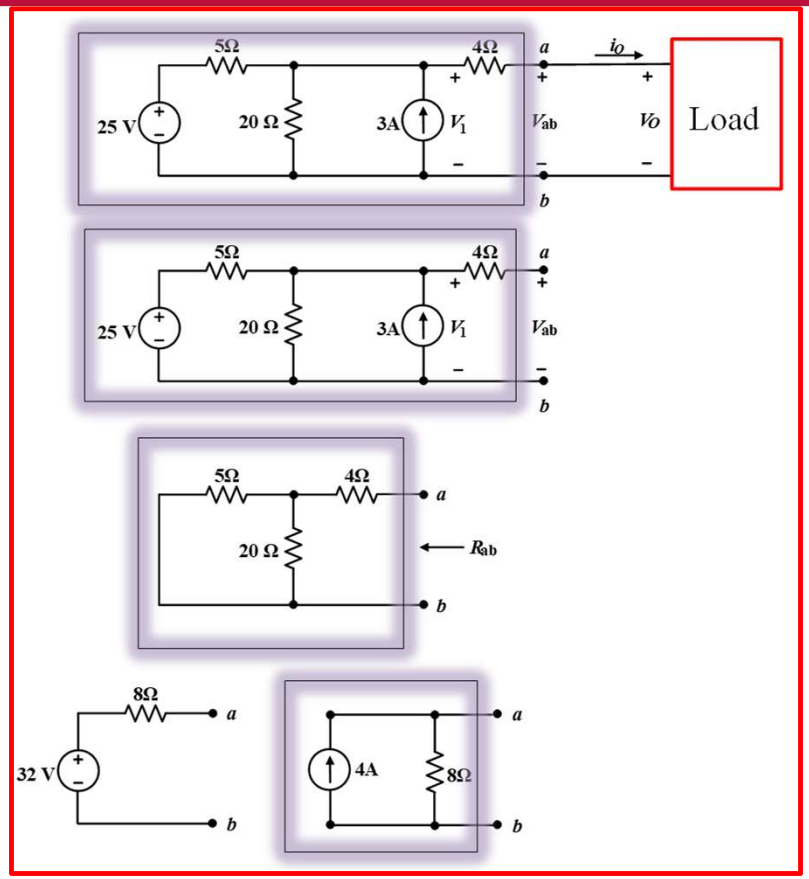
$$R_{Th} = \frac{V_{Th}}{i_{sc}} = \frac{v_{oc}}{i_{sc}}$$

Step 5: Calculate the Norton equivalent:

$$I_N = i_{sc}\,, \qquad R_N = R_{Th} = \frac{v_{oc}}{i_{sc}}$$

$$R_{Th} = \frac{32}{4} = 8\Omega$$

2 analyses Needed!



Networks with Only Independent Sources: Method 2→ eq Resistance + oc Voltage or sc Current

Step 1: Detach the network from the load (rest of the circuit).

Step 2: Find the oc voltage (or sc current) across the two detached network terminals (v_{oc}): $V_{Th} = v_{oc}$

Step 3: Deactivate all the **independent** current and voltage sources in the detached network:

- Voltage sources are short circuited.
- Current sources are open circuited.

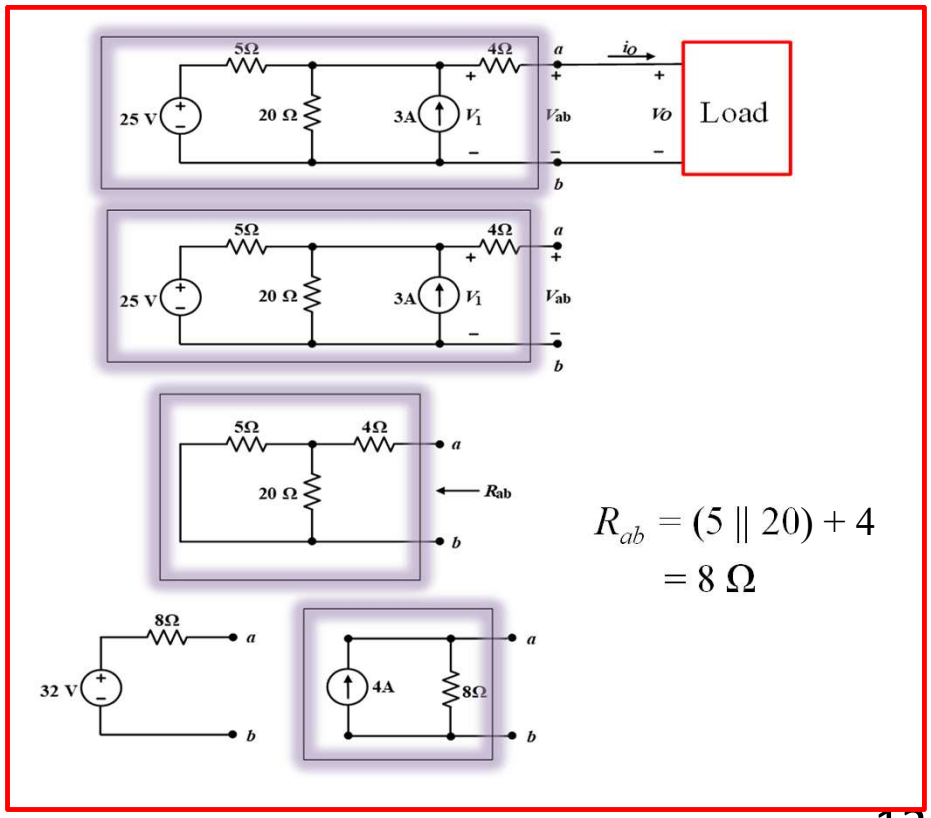
Step 4: Find the equivalent resistance from the two detached terminals:

$$R_{Th} = R_{eq} = R_{ab}$$

Step 5: Calculate the Norton equivalent:

$$I_N = i_{sc}\ ,\ R_N = R_{Th} = \frac{v_{oc}}{i_{sc}}$$

2 analyses Needed!



$$R_{ab} = (5 \parallel 20) + 4 = 8\ \Omega$$

Networks with Only Dependent Sources: Method 3→A Test Source

Step 1: Detach the network from the load (rest of the circuit).

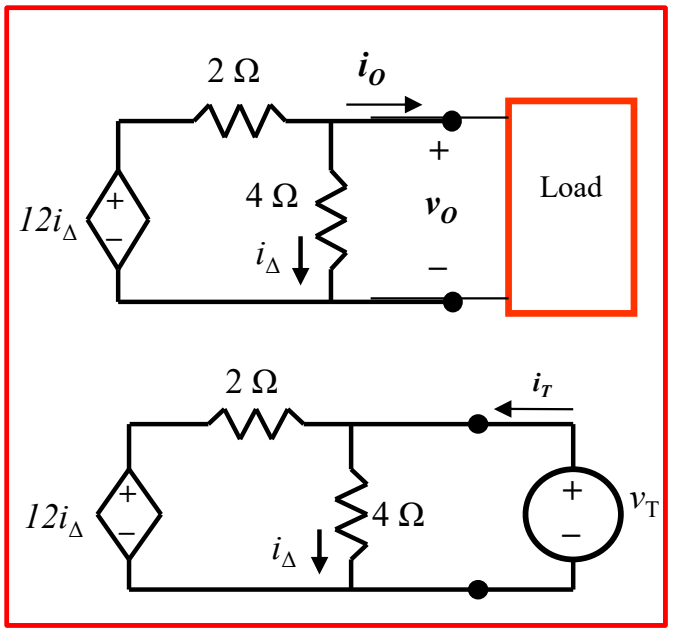
Step 2: Since there is no independent source of excitation in the circuit:
$v_{oc} = 0 \Rightarrow V_{Th} = 0$
Therefore, the Thévenin equivalent circuit is just the Thévenin resistor.

Step 3: The only way to find R_{Th} is by using test voltage and finding the terminal voltage and current v_T, i_T, and then:

$$R_{Th} = \frac{v_T}{i_T} \qquad V_T = 4i_\Delta = 2(i_T - i_\Delta) + 12i_\Delta = 2i_T + 10\frac{V_T}{4}$$

Step 4: Find the equivalent resistance from the two detached terminals:

$$R_{Th} = R_{eq} = R_{ab} \qquad \frac{3}{2}V_T = -2i_T \xrightarrow{\text{yields}} \frac{V_T}{i_T} = R_{Th} = -\frac{4}{3}\Omega$$



2 analyses Needed!

Networks with Both Dependent and Independent Sources: Method 1

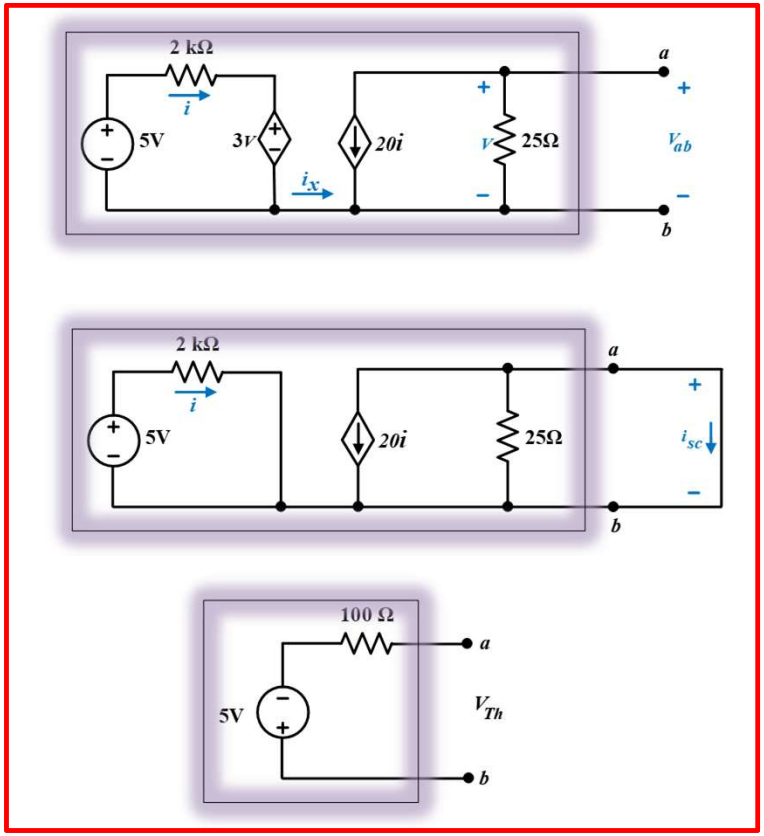
Step 1: Detach the network from the load (rest of the circuit).

Step 2: Find the open circuit voltage across the two detached network terminals (v_{oc}): $V_{Th} = v_{oc}$

Step 3: Short circuit the two detached network terminals and find the current in the short circuit wire (i_{sc}).

Step 4: Calculate the Thévenin resistance: $R_{Th} = \dfrac{V_{Th}}{i_{sc}} = \dfrac{v_{oc}}{i_{sc}}$

Step 5: Calculate the Norton equivalent.



Networks with Both Dependent and Independent Sources: Method 3

Step 1: Detach the network from the load (rest of the circuit).

Step 2: Find the open circuit voltage across the two detached network terminals (v_{oc}): $V_{Th} = v_{oc}$

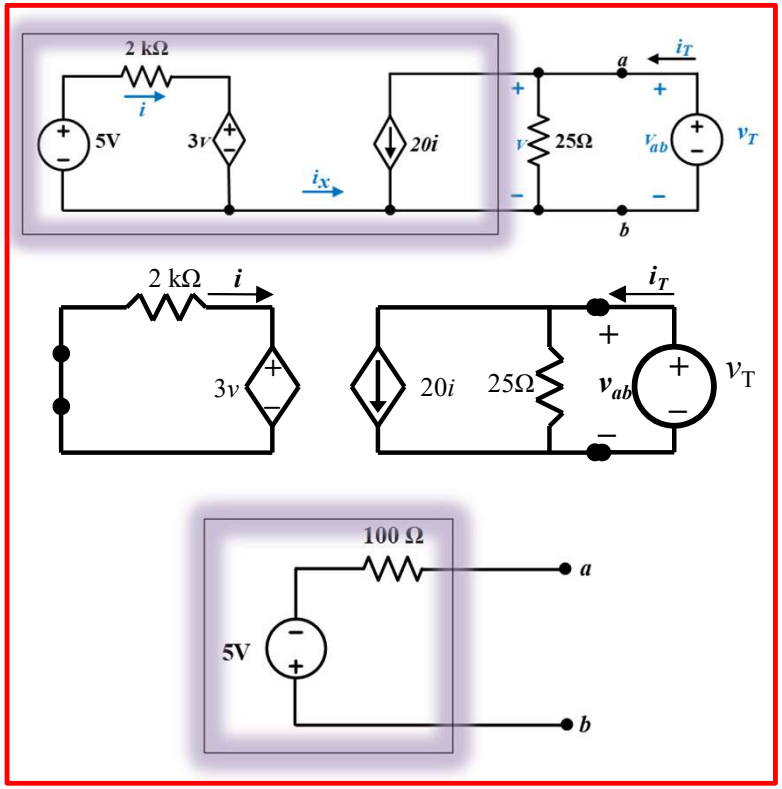
Step 3: Deactivate all the **independent** current and voltage sources in the detached network.

Step 4: The ONLY way to find R_{Th} is by using test voltage and finding the terminal voltage and current v_T, i_T, and then:

We find the equivalent resistance from the two detached terminals:

$$R_{Th} = R_{eq} = R_{ab}$$

$$R_{Th} = \frac{v_T}{i_T}$$



Networks with Both Dependent and Independent Sources: Method 4 (only one analysis needed)

Step 1: Detach the network from the load (rest of the circuit).

Step 2: Apply a test voltage or current source (*VT* or *IT*) to the network. Your choice must be based on the method of analysis you will adopt thereafter (nodal or mesh?).

Step 3: Analyze the circuit and find the other network **port** variable in terms of the test source value treated as a **parameter**. Then, we have one of the following:

Networks with Both Dependent and Independent Sources: Method 4 (only one analysis needed)

$$V_T = R_{eq} \cdot I_T + v_{oc}$$

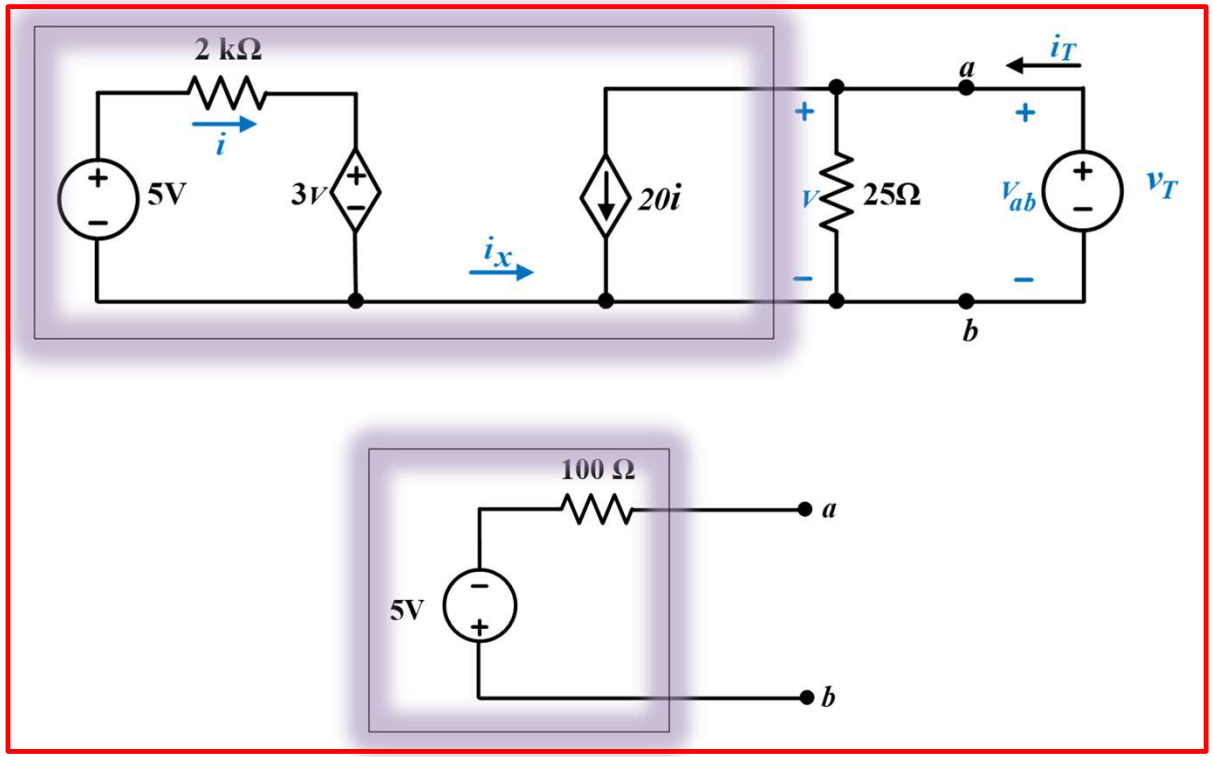
$$I_T = \frac{V_T}{R_{eq}} - I_{sc}$$

2 out of 3 quantities Found at once!

$$V_T = 25(i_T - 20i)$$

$$i = \frac{5 - 3V_T}{2000}$$

$$V_T = 100i_T - 5$$



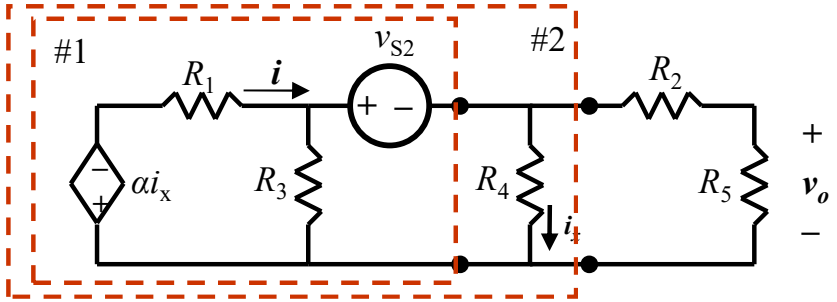
What Methods?

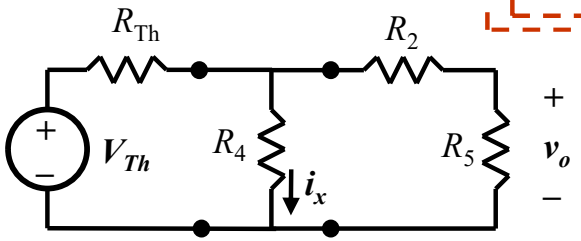
	Method 1 (v_{oc}, i_{sc})	**Method 2** (R_{eq}, v_{oc}) or (R_{eq}, i_{sc})	**Method 3** Test signal v_T or Test signal I_T	Equivalent Circuit
Independent sources only	✓	✓	✓	R_{Th}, V_{Th} R_N, I_N
Both independent and dependent sources	✓	✗	✓	R_{Th}, V_{Th} R_N, I_N
Dependent sources only	✗	✗	✓	R_{Th}, (V_{Th} = 0) R_N, (I_N = 0)
Method 4 is the Most general and Always works!	$V_{Th} = v_{oc}$ $I_N = i_{sc}$ $R_{th} = v_{oc}/i_{sc}$ $R_N = R_{Th}$	$V_{Th} = v_{oc}$ $I_N = i_{sc}$ $R_{th} = R_{eq}$ $R_N = R_{Th}$	$V_{Th} = v_{oc}$ $I_N = i_{sc}$ $R_{Th} = v_T/i_T$ $R_N = R_{Th}$	

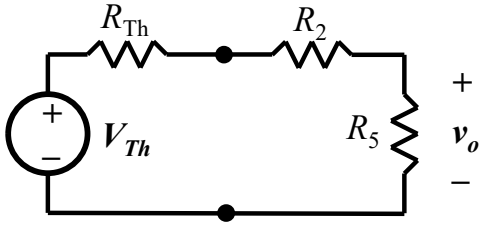
Warning: Networks with Dependent Sources

Make sure that the controlling current (e.g., $\boldsymbol{i}_x$) or voltage is in the same detached network with the dependent source.

$R_1 = R_2 = R_5 = 1\ \text{k}\Omega$
$R_3 = R_4 = 2\ \text{k}\Omega$
$\alpha = 2000$, $v_{S2} = 12$ V
$\boldsymbol{v}_o = ?$







Construction of the Thévenin equivalent circuit #1 is not possible since $\boldsymbol{i}_x$ is not part of the detached network.

#2 is okay:

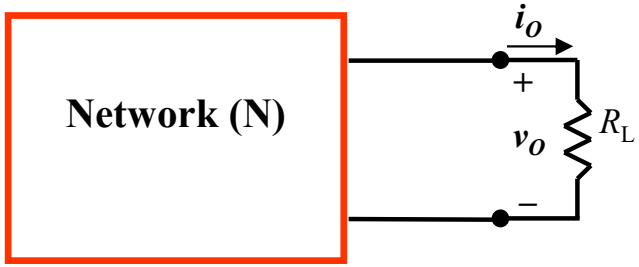
$V_{Th2} = (v_{cd})_{oc} = -6$ V
$i_{sc2} = (i_o)_{oc} = -18$ mA
$R_{Th2} = V_{Th2} / i_{sc2} = 1/3\ \text{k}\Omega$
$v_o = R5\ .V_{Th2}/(R_{Th2} + R_2 + R_5) = -18/7$ V

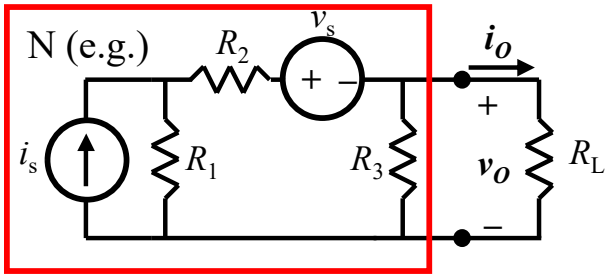
Maximum Power Transfer Theorem :MPTT

It is desired in many applications to maximize the power transmitted or delivered to a load.

Example: Maximum portion of the power generated by the power stations to our homes.

- For now, we consider maximum power transfer for a resistive circuit consisting of :
  - A resistive load R_L
  - A linear resistive network "N" with dependent and independent sources





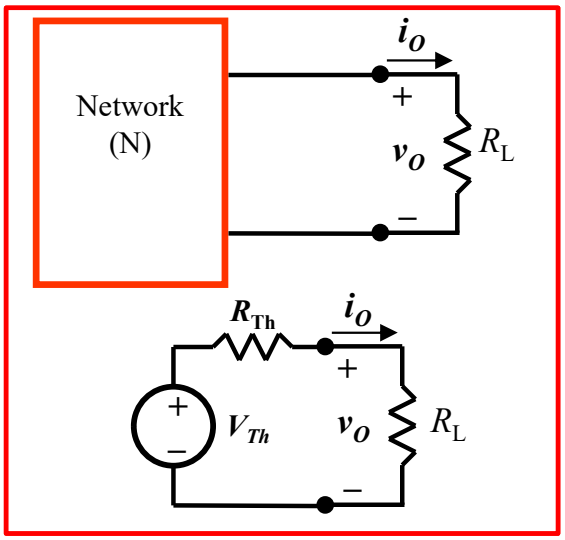
Problem Definition:

Find R_L that permits maximum power delivered to R_L?

Maximum Power Transfer Theorem (cont.)

Question: Find a suitable load $\boldsymbol{R_L}$ that permits the ***maximum*** power to be delivered from the network "N" to the load?

- $R_L = ?$
- $(p_L)_{max} = ?$



Answer: It is always possible to replace the network "N" with its Thévenin equivalent circuit:

$$p_L = R_L i_o^2$$

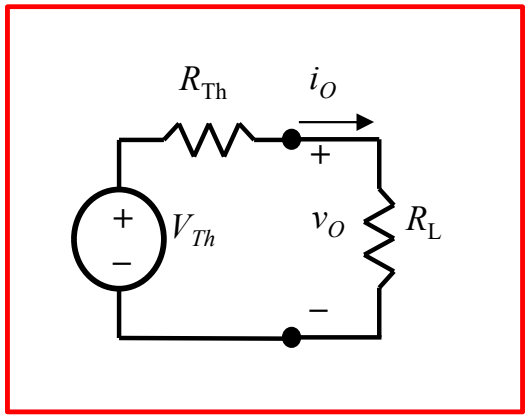
$$i_o = V_{Th}/(R_{Th} + R_L)$$

$$\Rightarrow p_L = \frac{R_L}{(R_{Th} + R_L)^2} V_{Th}^2$$

R_{Th} and V_{Th} depend on the network "N" characteristics and are fixed.

R_L will be adjusted to maximize p_L.

Maximum Power Transfer Theorem (cont.)



$$p_L = \frac{R_L}{(R_{Th}+R_L)^2} V_{Th}^2$$

We want to find the maximum:

$$\Rightarrow \frac{dp_L}{dR_L} = \left[\frac{(R_{Th}+R_L)^2 - 2R_L(R_{Th}+R_L)}{(R_{Th}+R_L)^4}\right] V_{Th}^2$$

$$\Rightarrow \frac{dp_L}{dR_L} = \left[\frac{(R_{Th}+R_L) - 2R_L}{(R_{Th}+R_L)^3}\right] V_{Th}^2$$

$$\frac{dp_L}{dR_L} = 0 \Rightarrow \left[\frac{(R_{Th}+R_L) - 2R_L}{(R_{Th}+R_L)^3}\right] V_{Th}^2 = 0 \qquad \Rightarrow R_L = R_{Th}$$

$$(p_L)_{\max} = p_L\big|_{R_L = R_{Th}} = \frac{R_{Th}}{(R_{Th}+R_{Th})^2} V_{Th}^2$$

$$\Rightarrow (p_L)_{\max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{V_{Th}^2}{4R_L}$$

Example: Maximum Power Transfer

- $(p_{RL})_{max}$ = ? :
- Step 1: Find the Thévenin equivalent circuit (v_{oc}, R_{Th})
- Open Circuit analysis (v_{oc}) (use mesh-current analysis):

$i_1 = i_{S1}$ (known)

$\mathbf{m_2}$: $v_{S2} + R_2(i_2 - i_1) + R_3 i_2 = 0$

$\Rightarrow (R_2 + R_3)\, i_2 = -v_{S2} + R_2 i_{S1}$

$\Rightarrow \; i_2 = (R_2 i_{S1} - v_{S2})/(R_2 + R_3)$

$\Rightarrow i_2 = 1/3$ mA

$v_{oc} = R_1 i_1 + R_3 i_2 \Rightarrow v_{oc} = 10$ V

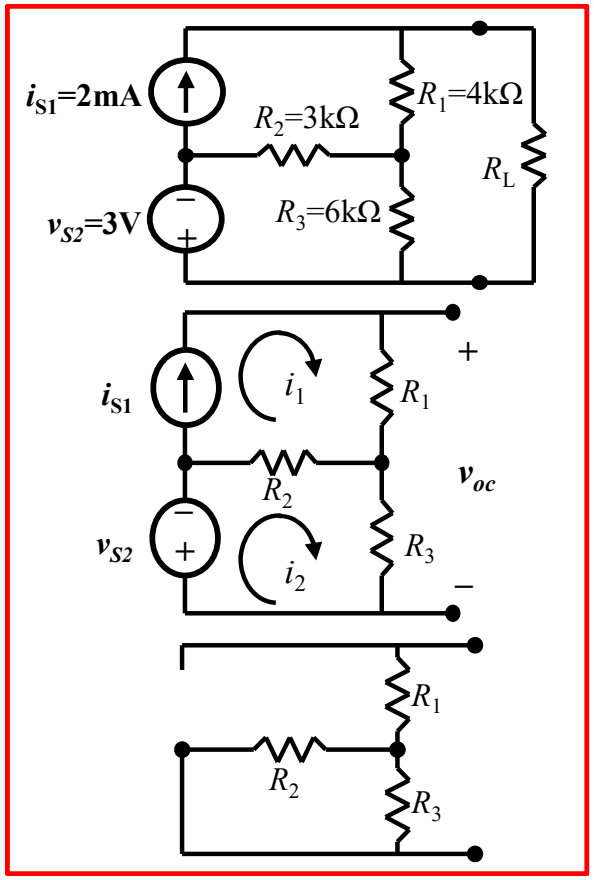
- Deactivate the circuit to find R_{Th}:

$R_{Th} = R_1 + (R_2 \,||\, R_3) \Rightarrow R_{Th} = 6\ \text{k}\Omega$

- Step 2: Compute the power

$(R_{RL})_{max} = R_{Th} = 6\ \text{k}\Omega$

$(p_{RL})_{max} = V_{Th}^2 / 4R_{Th} = 25/6$ mW



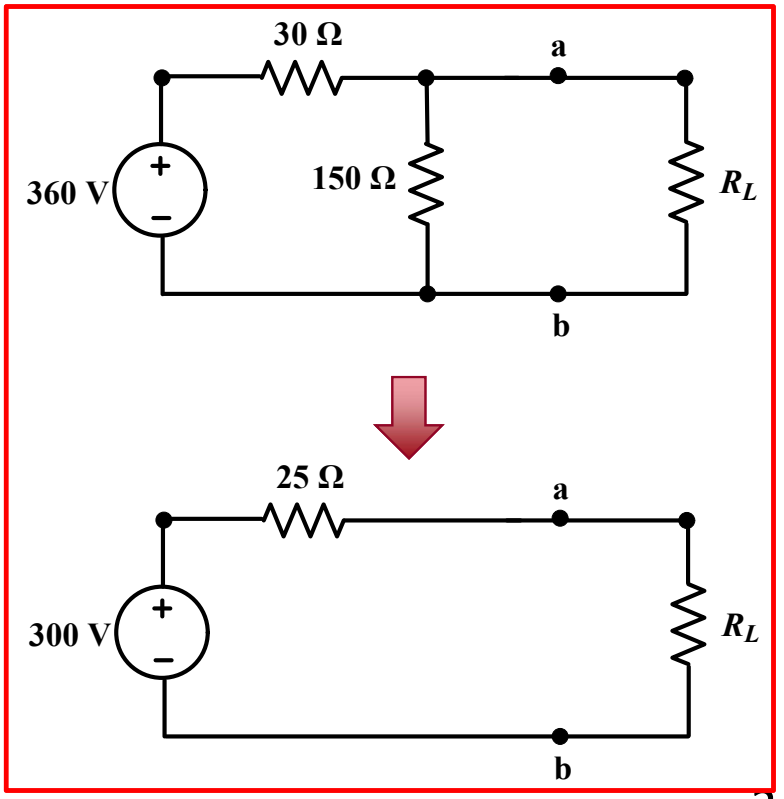
Example 2: Maximum Power Transfer

Determine:

a) The load R_L that will provide the maximum power transfer

b) The maximum power delivered to the load.

c) With this load, what percentage of power is transferred to the load?



Example 2: Maximum Power Transfer

a) Find the Thévenin equivalent:

$V_{Th} = v_{oc} = 150/180 \times 360 = 300$ V

- Using voltage divider
- *Or use node-voltage, etc.*

$R_{Th} = 150||30 = 25\ \Omega$

- Using Method 2 (or use 1)

b) Maximum power:

$p_{max} = 300^2/(4 \times 25) = 900$ W

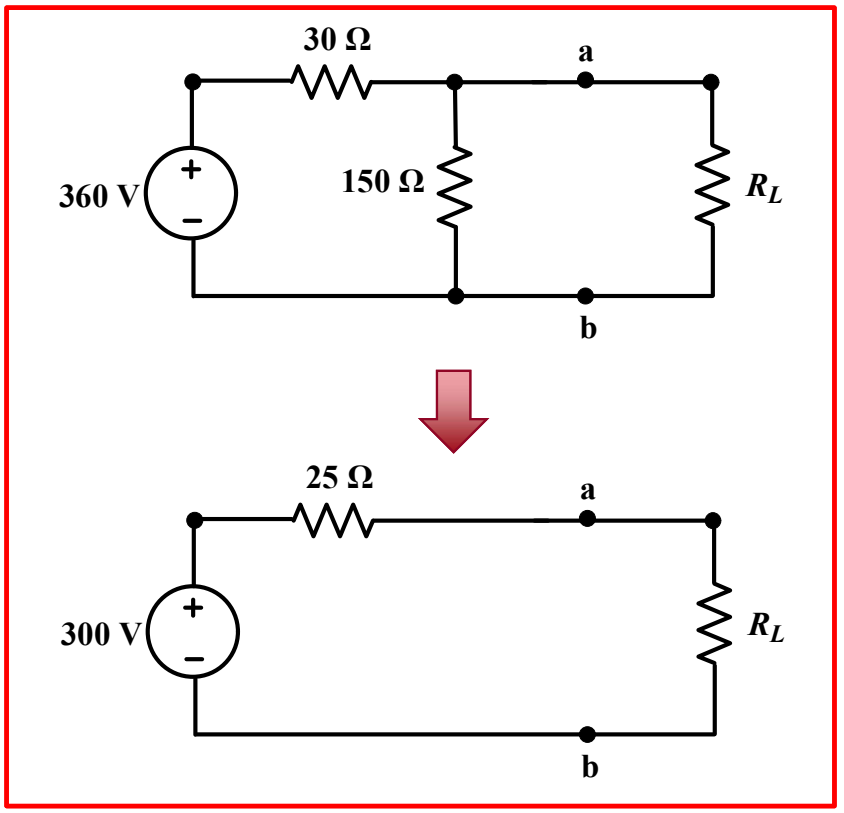
c) Percentage power delivered to load:

With $R_L = 25\ \Omega$, $v_{ab} = 150$ V

$\Rightarrow \quad p_{total} = 150^2/25 + 150^2/150 + 210^2/30$

$= 900 + 150 + 1470 = 2520$ W

=> Efficiency = 900/2520 = 35.7 %



Principle of Superposition (SWAT)

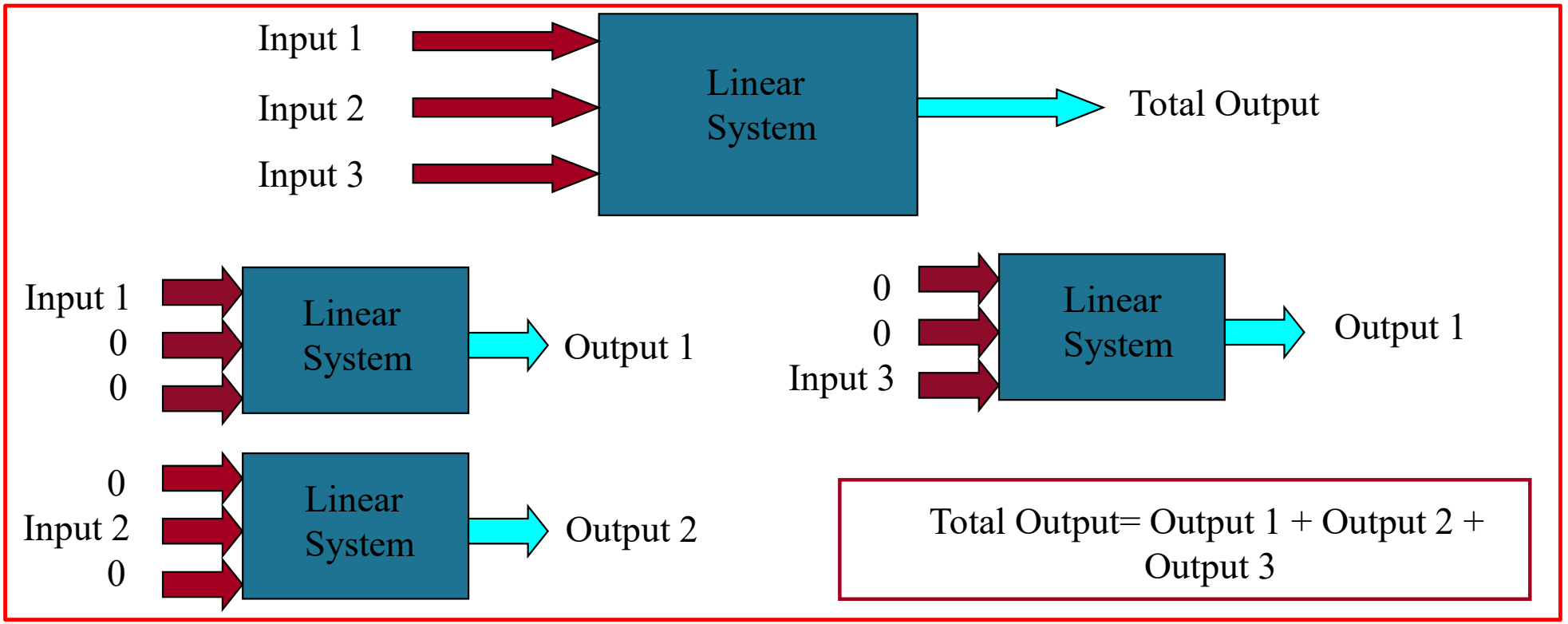
Every linear system obeys “the principle of superposition”.:

- **Principle of superposition:** when a system/circuit is excited by a number of (independent) sources or excitations, the total response/output = sum of the individual responses.

- Individual response: response of the system with only a single excitation (the rest are turned off).

Principle of Superposition (cont.)

Consider the multi-input single-output abstract system given below:



Principle of Superposition (cont.)

We restrict our discussion to resistive circuits but superposition applies to any linear circuit (including circuits with capacitors and inductors).

Note: the inputs or excitations in circuits are only "independent sources" .

Reminder: dependent or controlled sources are not really sources: they are mathematical models for more complex elements.

Principle of Superposition (cont.)

Example: Find i_o and v_o

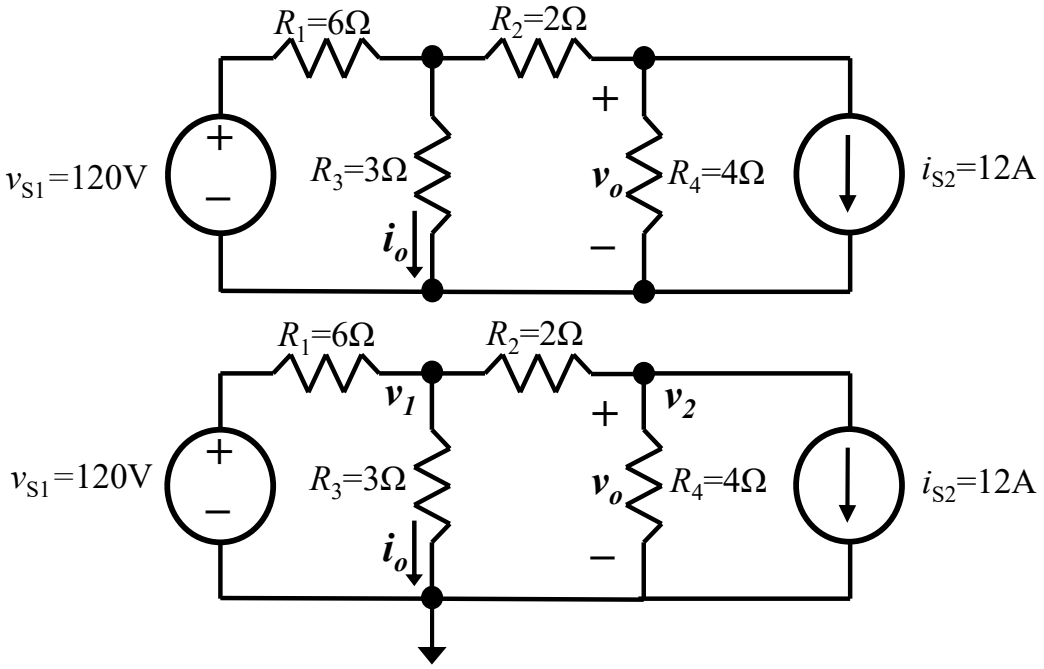
Using node-voltage:

$$\mathbf{n_1}: \quad \frac{v_1 - v_{S1}}{R_1} + \frac{v_1}{R_3} + \frac{v_1 - v_2}{R_2} = 0$$

$$\mathbf{n_2}: \quad \frac{v_2 - v_1}{R_2} + \frac{v_2}{R_4} + i_{S2} = 0$$

$$\Rightarrow \begin{cases} v_1 - \dfrac{v_2}{2} = \dfrac{v_{S1}}{6} \\ \dfrac{-v_1}{2} + \dfrac{3}{4} v_2 = -i_{S2} \end{cases} \Rightarrow \begin{cases} v_2 = \dfrac{v_{S1}}{6} - 2i_{S2} \\ v_1 = \dfrac{v_{S1}}{4} - i_{S2} \end{cases}$$

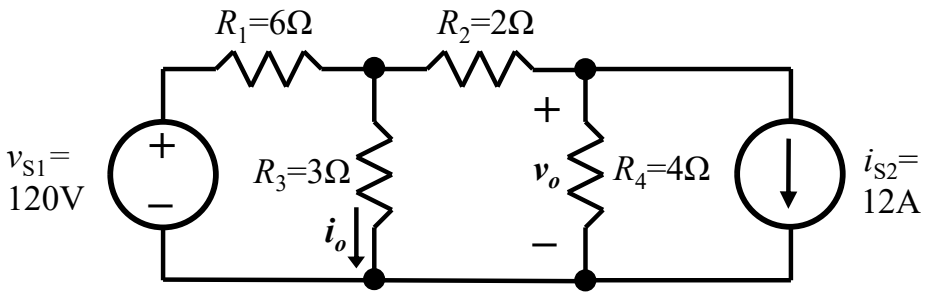
$$\Rightarrow v_0 = \frac{v_{S1}}{6} - 2i_{S2} \qquad \Rightarrow i_0 = \frac{v_{S1}}{12} - \frac{i_{S2}}{3}$$



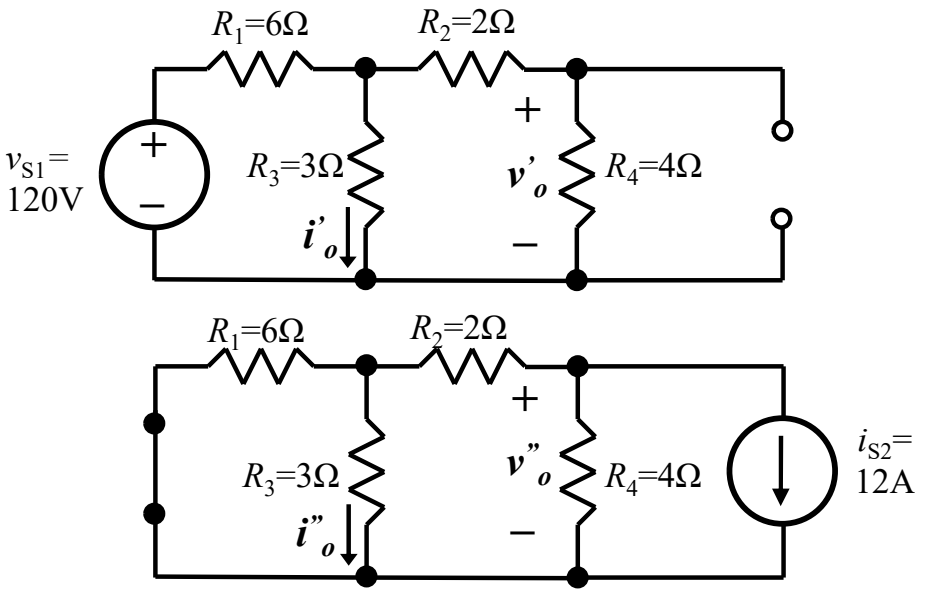
Note: The system responses i_o and v_o are combinations of the contributions of each of the independent sources v_{S1} and i_{S2}.

Principle of Superposition (cont.)

In any linear circuit containing multiple independent sources, the current or voltage of any point in the circuit may be calculated as the algebraic sum of the individual contributions of each independent source acting alone.



$$v_0 = v'_0 + v''_0$$
$$i_0 = i'_0 + i''_0$$



Principle of Superposition (cont.)

Acting alone?

- In each analysis, turn the other sources or excitations off.
- In circuits:

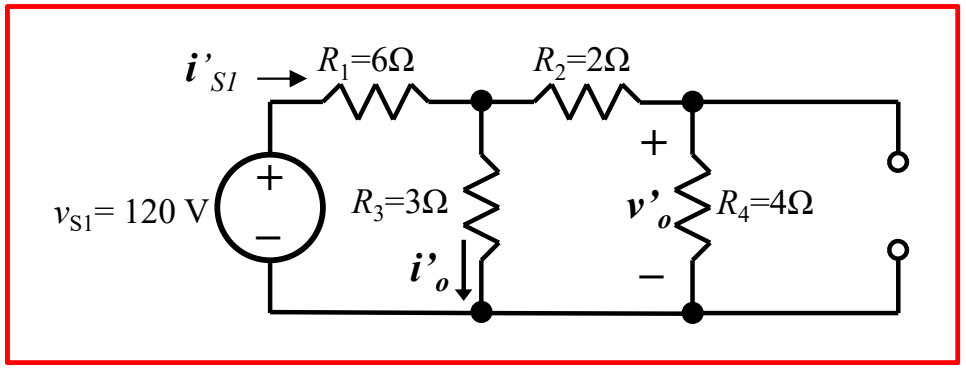
  - Current source (independent) off ⇨ open it.

  - Voltage source)independent) off ⇨ short it.

Principle of Superposition (cont.)

Investigate the principle of superposition in the previous example.

1) Deactivate i_{S2} ($i_{S2} = 0$):



$$i'_{S1} = \frac{v_{S1}}{[R_1 + ((R_2 + R_4) \,||\, R_3)]} = \frac{v_{S1}}{8}$$

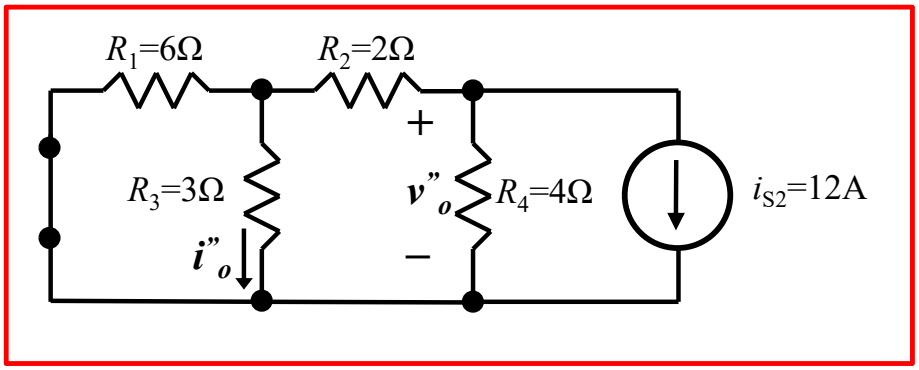
Current divider:

$$i'_0 = \frac{(R_2 + R_4)}{(R_2 + R_4) + R_3} i'_{S1} = \frac{v_{S1}}{12}$$

$$v'_0 = R_4 (i'_{S1} - i'_0) = \frac{v_{S1}}{6}$$

Principle of Superposition (cont.)

2) Deactivate v_{S2} ($v_{S2} = 0$):



$$v_0'' = -i_{S2}\left[\left[(R_1 \parallel R_3) + R_2\right] \parallel R_4\right] = -2i_{S2}$$

Current divider:

$$i_{R2} = \frac{R_4}{\left[R_4 + (R_2 + R_1) \parallel R_3\right]} i_{S2} = +\frac{i_{S2}}{2}$$

$$i_0'' = \frac{-R_1}{R_1 + R_3} i_{R2} = \frac{-i_{S2}}{3}$$

$$v_0 = v'_0 + v''_0 = v_{S1}/12 - i_{S2}/3$$
$$i_0 = i'_0 + i''_0 = v_{S1}/6 - 2i_{S2}$$

Superposition and Dependent Sources

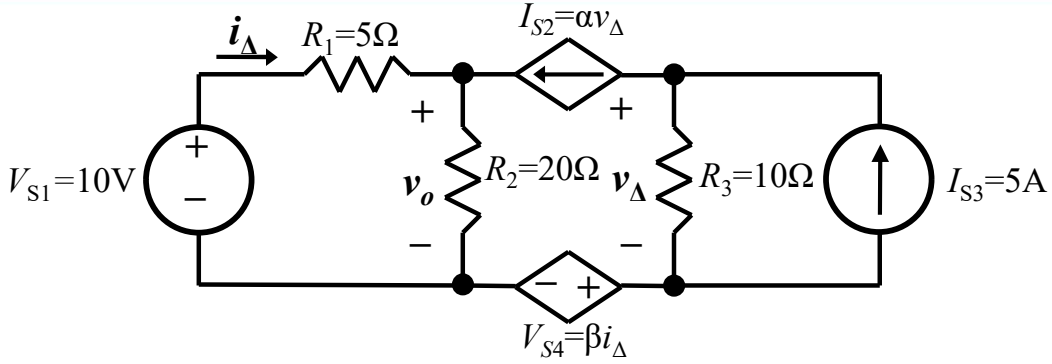
Note: Deactivate **only** the independent sources to apply superposition.

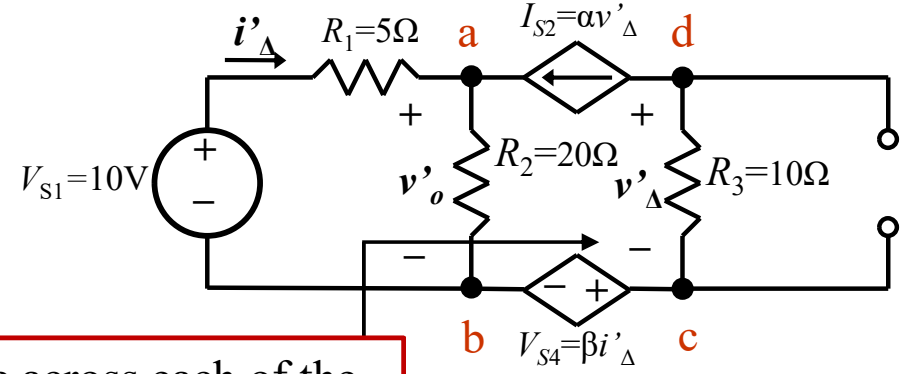
We have: $\beta = 2$, $\alpha = 0.4$.

Find v_0 .

Deactivate I_{S3} ($I_{S3} = 0$):

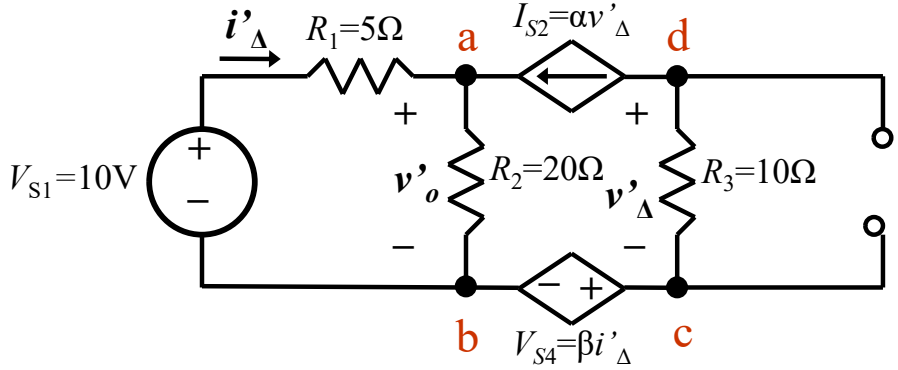
$$v'_\Delta = -R_3 I_{S2} = -\alpha R_3 v'_\Delta$$
$$\Rightarrow (1 + \alpha R_3) v'_\Delta = 0$$
$$\Rightarrow 5v'_\Delta = 0$$
$$\Rightarrow v'_\Delta = 0$$
$$\Rightarrow I_{S2} = 0$$





Can you find the voltage across each of the elements in the mesh abcda in this case?

Superposition and Dependent Sources

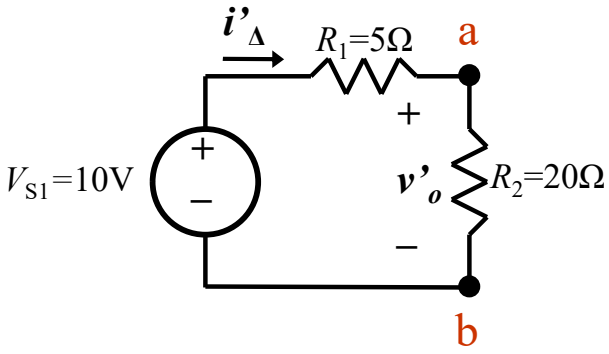


Deactivate I_{S3} ($I_{S3} = 0$):

$$v'_\Delta = -R_3 I_{S2} = -\alpha R_3 v'_\Delta$$
$$\Rightarrow (1 + \alpha R_3)\, v'_\Delta = 0$$
$$\Rightarrow 5v'_\Delta = 0$$
$$\Rightarrow v'_\Delta = 0$$
$$\Rightarrow I_{S2} = 0$$

Open circuit at terminal **a-b**

$$v_0' = \frac{R_2}{R_1 + R_2} V_{S1} \Rightarrow v_0' = 8\ \text{V}$$



Superposition and Dependent Sources

Deactivate V_{S1} ($V_{S1} = 0$):

Use node-voltage analysis to find v''_0

$$\mathbf{n_1}: \frac{v_1}{R_1} + \frac{v_1}{R_2} - I_{S2} = 0$$

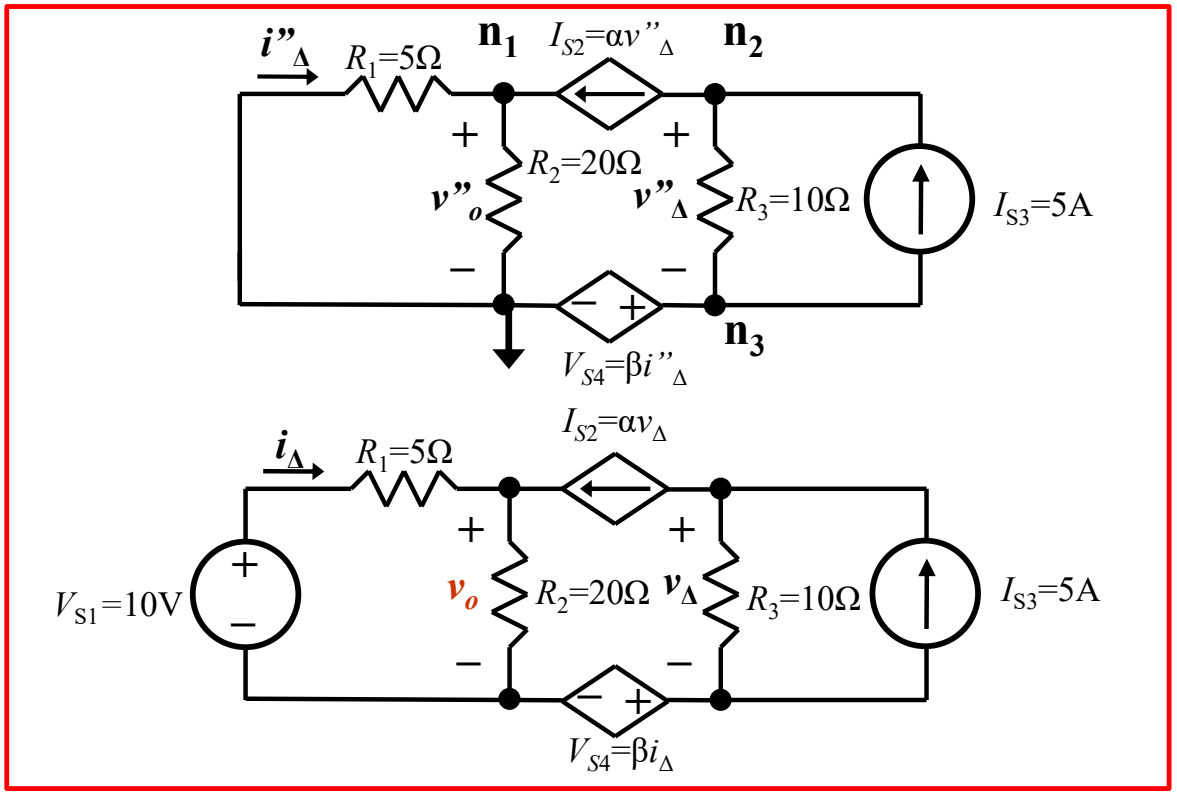
$$\mathbf{n}_2: \frac{v_2 - v_3}{R_3} - I_{S3} + I_{S2} = 0$$

$v_3 = v_{S4} = \beta i''_\Delta = \beta(-v_1)/R_1$

$I_{S2} = \alpha v''_\Delta = \alpha(v_2 - v_3) = \alpha(v_2 + \beta v_1/R_1)$

$\Rightarrow v''_0 = v_1 = 16$ V

$$v_0 = v'_0 + v''_0 = 8 + 16 = 24 \text{ V}$$



Superposition and Power

WARNING! Superposition **can NOT** be used to calculate **power** in an element since power is a nonlinear function.

- ❖ Power in an element is **NOT** equal to the sum of the powers calculated in the element by considering one independent source at a time.

$$v = v' + v'' \quad , \quad i = i' + i'' \qquad \text{(superposition)}$$
$$p = vi = (v' + v'')(i' + i'') = (v'i' + v'i'') + (v''i' + v''i'')$$
$$p' = v'i' \ , \ p'' = v''i''$$
$$\Rightarrow \ p \neq p' + p''$$

$\Rightarrow$ **Superposition is not valid for power**

Readings

❖Chapter 3 is done.

❖Do all the Assessment problems and check with the solutions posted on moodle and ME.

❖Go through the study guide for the chapter.

❖Practice with the workbook on ME.